INCH-POUND

MIL-DTL-16104C
7 October 2026
SUPERSEDING
MIL-S-16104B
27 December 1983

DETAIL SPECIFICATION

SYNCHROSCOPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers switchboard-mounted synchroscopes that indicate phase angle between two alternating current (AC) power sources or show within limits their relative departure from synchronism (see 6.5).

1.2 Classification. The type designations of synchroscopes are in the following form as specified (see 6.2).

MR	4R	W	060	SY	H
Component	Style	Color scheme	Frequency response	Unit	Special features
1.2.1	1.2.2	1.2.3	1.2.4	1.2.5	1.2.6

1.2.1 Component. Synchroscopes covered by this specification are designated by the two-letter symbol “MR”.

1.2.2 Style. The style is designated by two characters that signify the size and shape as follows (see figures 1 and 2).

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AMSC N/A FSC 6625

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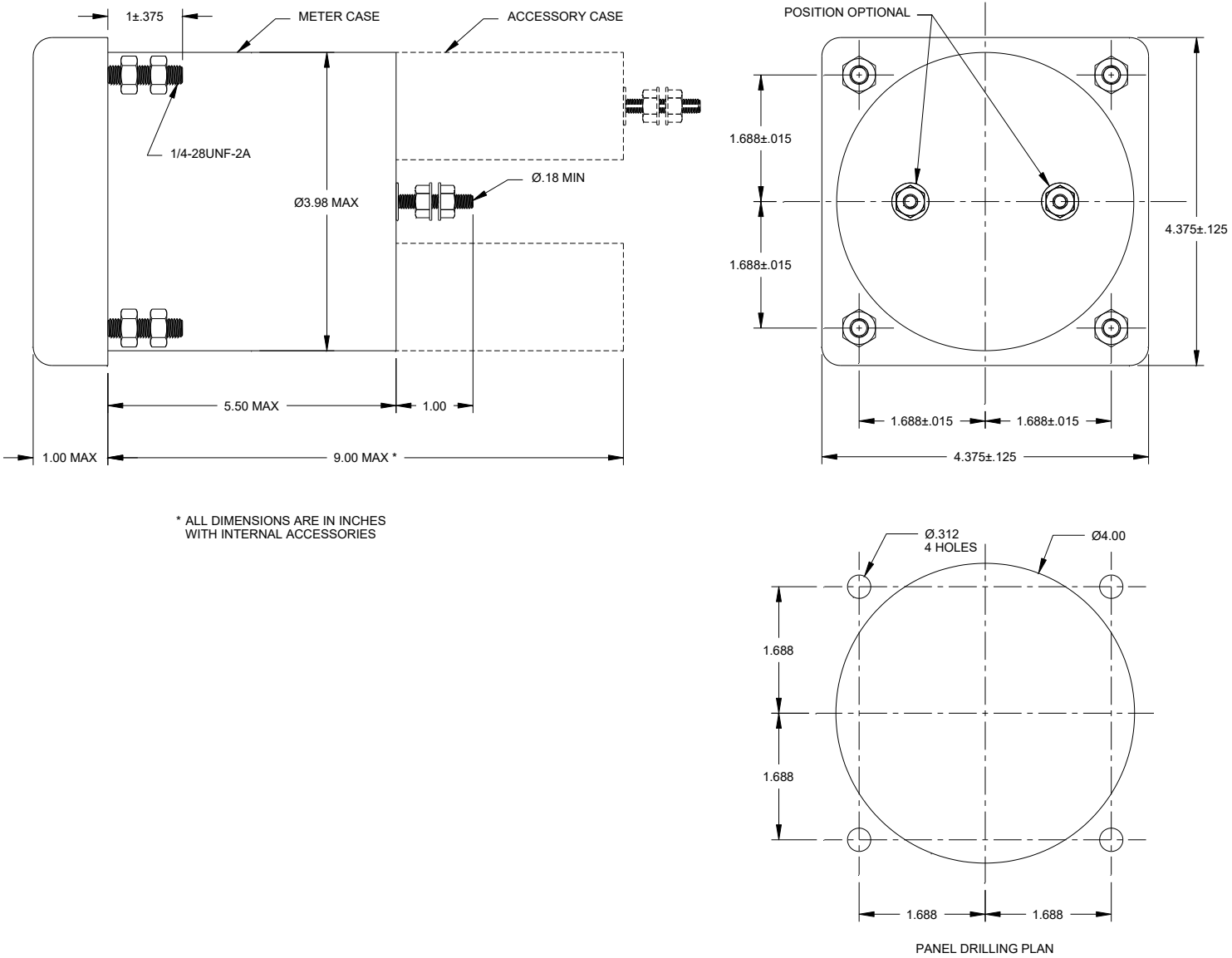


FIGURE 1. Rectangular meter (4½ inches).

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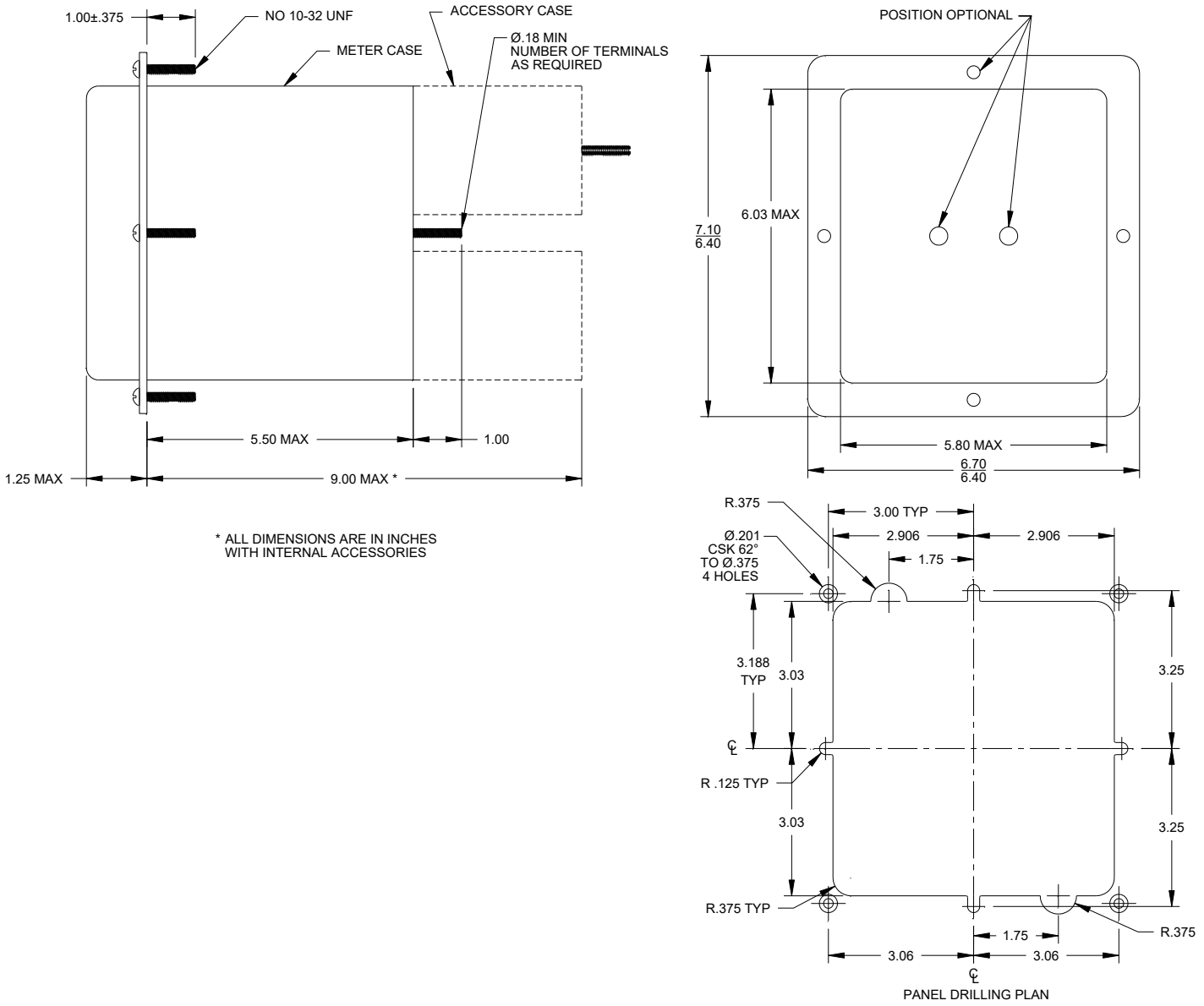


FIGURE 2. Rectangular meter (6 inches).

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1.2.2.1 First digit.

4: 4½ -inch synchroscope

6: 6-inch synchroscope

1.2.2.2 Second letter.

R: Round flange

S: Rectangular flange

1.2.3 Color scheme.

B: Black dial, white pointer and lettering

W: White dial, black pointer and lettering

1.2.4 Frequency response. The frequency response is designated by a three-digit number in hertz (Hz), zero-filled to the left.

1.2.5 Unit. The unit is "SY" for synchroscope.

1.2.6 Special features. (when applicable)

H: High-impact (HI) shock

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-15090	-	Enamel, Equipment, Light Gray (Navy Formula No. 111)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this publication are available online at https://asq.org.)

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ASTM INTERNATIONAL

ASTM D257 - Standard Test Methods for DC Resistance or Conductance of Insulating Materials

ASTM D709 - Standard Specification for Laminated Thermosetting Materials

ASTM D3935 - Standard Classification System and Basis for Specification for Polycarbonate (PC) Unfilled and Reinforced Material

ASTM D5948 - Standard Specification for Molding Compounds, Thermosetting

(Copies of these publications are available online at www.astm.org.)

SAE INTERNATIONAL

SAE J3097/ANSI Z26.1 - Standard for Safety Glazing Materials for Glazing Motor Vehicles and Motor Vehicle Equipment Operating on Land Highways – Safety Standard

(Copies of this publication are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Safety. The safety requirements shall be in accordance with the requirements of MIL-DTL-917.

3.3 Material. Synchroscopes shall be made of the material specified herein. However, when no definite material is specified, a material that will enable the synchroscopes to meet the performance requirements of this specification shall be used. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.3.1 Restricted materials. Material restricted for Navy use by MIL-DTL-917 shall not be used in the construction of synchroscopes.

3.3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.3 Plastics. When applicable, the following types of plastics shall be used for construction of synchroscopes:

a. Molded thermosetting plastic shall be in accordance with ASTM D5948.

b. Laminated thermosetting plastic shall be in accordance with ASTM D709.

c. Plastic windows shall be clear and non-toxic and shall possess antielectrostatic properties. Polycarbonate materials shall be in accordance with ASTM D3935. Antielectrostatic properties shall be tested in accordance with ASTM D257.

3.3.4 Metals. Metals and the treatment of metals shall be corrosion-resistant in accordance with MIL-DTL-917.

3.3.5 Glass. Glass shall be shatterproof-type, clear, and colorless in accordance with the SAE J3097/ANSI Z26.1 definition of laminated glass.

3.4 Operating voltage and frequency. The synchroscope shall operate with AC voltage between 105 and 125 volts (V) at the specified frequency under the reference conditions specified in 4.2.1.

3.5 Sensitivity. The synchroscopes shall be sensitive to a difference of frequency between two AC sources up to 1.5 Hz and shall be within 1 percent accuracy up to 0.1 Hz.

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3.6 Cases. Synchroscope cases shall be constructed only from the materials specified herein. The cases shall be constructed with close-fitting joints to minimize the entrance of dust and moisture. Metal cases shall be protected against corrosion in accordance with MIL-DTL-917.

3.7 Windows. Synchroscopes shall be provided with a glass or plastic window. Regardless of the type, it shall be free from discoloration, scratches, striae, or other defects that may interfere in any way with the normal reading of the indicator in daylight.

3.8 Dial and pointer.

3.8.1 Dial. The dial shall be readily removable and made of stiff material firmly secured to the mounting. The material and construction shall ensure freedom from warping, fading, or other deterioration during the tests specified in section 4. Paper dials will be acceptable only when adequately bonded to a metal support.

3.8.1.1 Scale markings. Scale markings shall indicate when one AC source is synchronous with the one to be synchronized and the speed of the incoming source to the one with which it is being synchronized. The words "slow" and "fast" shall indicate the speed of the incoming source with the dial pointer rotating in a counterclockwise direction for the former and a clockwise direction for the latter.

3.8.2 Pointer. The pointer shall be rigidly constructed and shall not become damaged or distorted when subjected to any of the tests of this specification.

3.9 Finish. The portion of the case exposed to view from the front of the panel shall have a light gray finish in accordance with formula no. 111 of MIL-DTL-15090. Light reflecting trimmings shall not be used. Metal cases shall be rendered suitably resistant to corrosion before the application of the final gray finish.

3.10 Identification plates. Identification and other marking plates shall be in accordance with MIL-STD-130.

3.11 Dielectric strength. Synchroscopes shall withstand the dielectric test specified in 4.5.2 without damage or flashover. A test voltage of 1900 V root mean square (rms) shall be applied as follows:

a. External live parts to external shield of the case.

b. External live parts to exposed metal parts on the front of indicator.

c. External live parts to metal mounting panel.

3.12 Insulation resistance. When tested as specified in 4.5.3, the insulation resistance of synchroscopes shall be at least 20 megohms.

3.13 Vibration. Synchroscopes shall withstand the vibration test specified in 4.5.4 without permanent change in indication greater than 1 percent of total deflection.

3.14 Moisture resistance. Synchroscopes shall operate freely following exposure to the moisture resistance test specified in 4.5.5.

3.15 Mounting. Mounting shall be in accordance with figures 1 or 2 as applicable. Attaching hardware shall be supplied.

3.16 Shock. When required (see 6.2), the synchroscopes shall be constructed to withstand the shock test specified in 4.5.6.

3.17 Voltage influence. A 7-percent voltage variation on both elements shall not produce a change of indication. A 1-percent voltage difference shall not change the indication by more than 2 degrees (see 4.5.7).

3.18 Overvoltage. The synchroscopes shall sustain the overvoltage test specified in 4.5.8.

3.19 Life cycle. The synchroscopes shall be made to withstand the life cycle test specified in 4.5.9. The synchroscope shall not overheat or change accuracy beyond 1 percent (3½ degrees).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

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4.2 Inspection conditions.

4.2.1 Reference conditions. All inspections shall be performed in accordance with the reference conditions specified in 3.4 and as follows:

a. Temperature: 73.4±3.6 °F (23±2 °C).

b. Relative humidity: Greater than 50 percent.

4.2.2 Thermal stability. Unless otherwise specified herein, synchroscopes shall be maintained at the specified inspection temperature for not less than 2 hours to attain thermal stability before taking readings.

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table I.

TABLE I. First article and conformance inspections.

Inspection	First article inspection	Conformance inspection	Requirement paragraph	Inspection paragraph
Visual inspection	X	X	3.3, 3.6 through 3.10, and 3.15	4.4.3
Operating voltage and frequency	X	X	3.4	4.5.1
Sensitivity	X	X	3.5	4.5.1
Dielectric strength	X		3.11	4.5.2
Insulation resistance	X	X	3.12	4.5.3
Vibration	X		3.13	4.5.4
Moisture resistance	X		3.14	4.5.5
Shock	X		3.16	4.5.6
Voltage influence	X		3.17	4.5.7
Overvoltage	X		3.18	4.5.8
Life cycle	X		3.19	4.5.9

4.4 Conformance inspection. Each production synchroscope shall be subjected to the conformance inspections specified in table I.

4.4.1 Inspection lot. All synchroscopes of the same type presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.4.2 Inspection sample. The inspection sample shall be the single sampling plan for normal inspection in accordance with ASQ/ANSI Z1.4 at inspection level II and the Acceptance Quality Level (AQL) specified (see 6.2).

4.4.3 Visual inspection. Each of the synchroscopes shall be examined to verify compliance with this specification. Any synchroscope containing one or more visual defects shall be rejected.

4.5 Methods of inspection.

4.5.1 Operating voltage and frequency and sensitivity. Each synchroscope shall be tested for compliance with 3.5 by connecting it to two AC sources (see 3.4), one of which shall be slowly increased until the synchroscope ceases to operate and then slowly decreased in the opposite direction until the synchroscope again ceases to operate. The synchroscope shall then be slowly returned to 0 frequency difference. This test shall be repeated with the inputs interchanged.

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4.5.2 Dielectric strength. Synchroscopes shall be mounted on steel panels by the mounting screws furnished with the indicators. The synchroscopes shall be subjected to a voltage of the rms value specified in 3.11 applied between all external live parts of the indicator and the following:

a. The external shield of the case.

b. Exposed metal parts on the front of the synchroscope.

c. The metal panel on which the synchroscope is mounted.

The frequency shall be nominal 60.0 Hz and shall approximate as closely as possible a true sine wave in form. The voltage shall be applied continuously for 60 seconds. The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage directly from the circuit or indirectly via instrument transformer. To avoid voltage surges, the applied voltage shall be raised to its full value gradually and, upon completion of the 60-second test, gradually reduced.

4.5.3 Insulation resistance. The synchroscope shall be mounted on a steel panel. Using 500 V direct current, the insulation resistance (see 3.12) shall be determined between all external live parts and the following:

a. The external shield of the case.

b. Exposed parts on the front of the synchroscope.

c. The metal panel on which the synchroscope is mounted.

4.5.4 Vibration. Synchroscopes shall be subjected to the type I tests of MIL-STD-167-1. The changes in indication shall then be determined (see 3.13).

4.5.5 Moisture resistance. The synchroscope shall be placed in a humidity chamber and kept for 6 hours at a temperature of 149 °F (65 °C) and a relative humidity of 90 percent. Electrical energy shall be supplied to the synchroscope. At the end of the 6-hour period, the heat shall be shut off and the meter deenergized. The synchroscope shall then be permitted to cool for 18 hours in this atmosphere, the relative humidity of which rises as the temperature declines. (Precaution shall be taken to prevent dripping on the synchroscope.) A temperature of 95 °F (35 °C) or lower, but not less than 77 °F (25 °C), shall be attained during the cooling period. The complete cycle shall then be repeated. Upon completion of the second cycle, the synchroscope shall be removed from the humidity chamber and permitted to dry for 24 hours at room temperature with the relative humidity less than 60 percent. Synchroscope readings shall then be taken at the reference temperature. The meter window shall be removed, and the synchroscope shall then be subjected to two additional cycles. The synchroscope shall be removed from the humidity chamber and permitted to dry for 24 hours at room temperature with the relative humidity less than 60 percent. Meter readings shall then be taken at reference temperature. Synchroscope readings shall be within specified tolerances. Only one synchroscope of the sample shall be subjected to this test (see 3.14).

4.5.6 Shock (HI shock type only). One synchroscope shall be tested in accordance with the grade A, class I, type B shock test specified in MIL-DTL-901. The features of the test shall be as follows:

a. Weight designation of the shock test: Lightweight.

b. Definition of failure to perform principal functions:

(1) Loosening, cracking, or damaging of any screws, bearings, pivots, transparent cover, or other parts.

(2) Appreciable distortion of any parts.

(3) Inability to indicate synchronism.

(4) Impairment of specified sensitivity.

c. Acceptable method of mounting the equipment on the shock-testing machine: In accordance with MIL-DTL-901.

d. Place at which shock test will be conducted: At a Government laboratory specified by the command or agency concerned.

4.5.7 Voltage influence. Each synchroscope shall be tested for voltage influence as specified in 3.17 and meet the requirements specified therein.

4.5.8 Overvoltage. Each synchroscope shall be energized at nominal voltage. The frequencies shall be mismatched to cause a rotation of one revolution per 10 to 20 seconds. The voltage shall be set on both elements at

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125 V and then one element raised at 170 V and held for 2 minutes. The synchroscope shall be deenergized and the test repeated, energizing the other element to 170 V. The meter shall then be subjected to the conformance tests (see 3.18).

4.5.9 Life cycle. Each synchroscope shall be cycled for 720 hours, approximately 4 minutes on and 1 minute off. The speed of rotation shall be a minimum of one revolution per 10 seconds. The indicator shall be monitored during one cycle of each 24-hour period of operation and perform as outlined in 3.19.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Synchroscopes covered by this specification are intended for combatant naval shipboard use and in general application of electrical measurement, including land installation. High-impact shock synchroscopes are normally used in combatant naval ships. Current stock of synchroscopes may be used until depleted; future requisitions for synchroscopes should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. When first article inspection is required (see 3.1).

d. When synchroscopes are required to withstand shock testing (see 3.16).

e. AQL of 1.0 (see 4.4.2).

f. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item, and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that the bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitations.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Synchronism. Synchronism is the condition existing between two or more periodic quantities of the same period when the phase difference between them is zero.

6.6 Subject term (key word) listing.

Alternating current
Frequency
Indicator, electrical
Meter, electrical

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Shock resistance
Switchboard

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH

Preparing activity:
Navy – SH
(Project 6625-2025-003)

Review activities:
Army – CR4
Navy – AS, MC

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